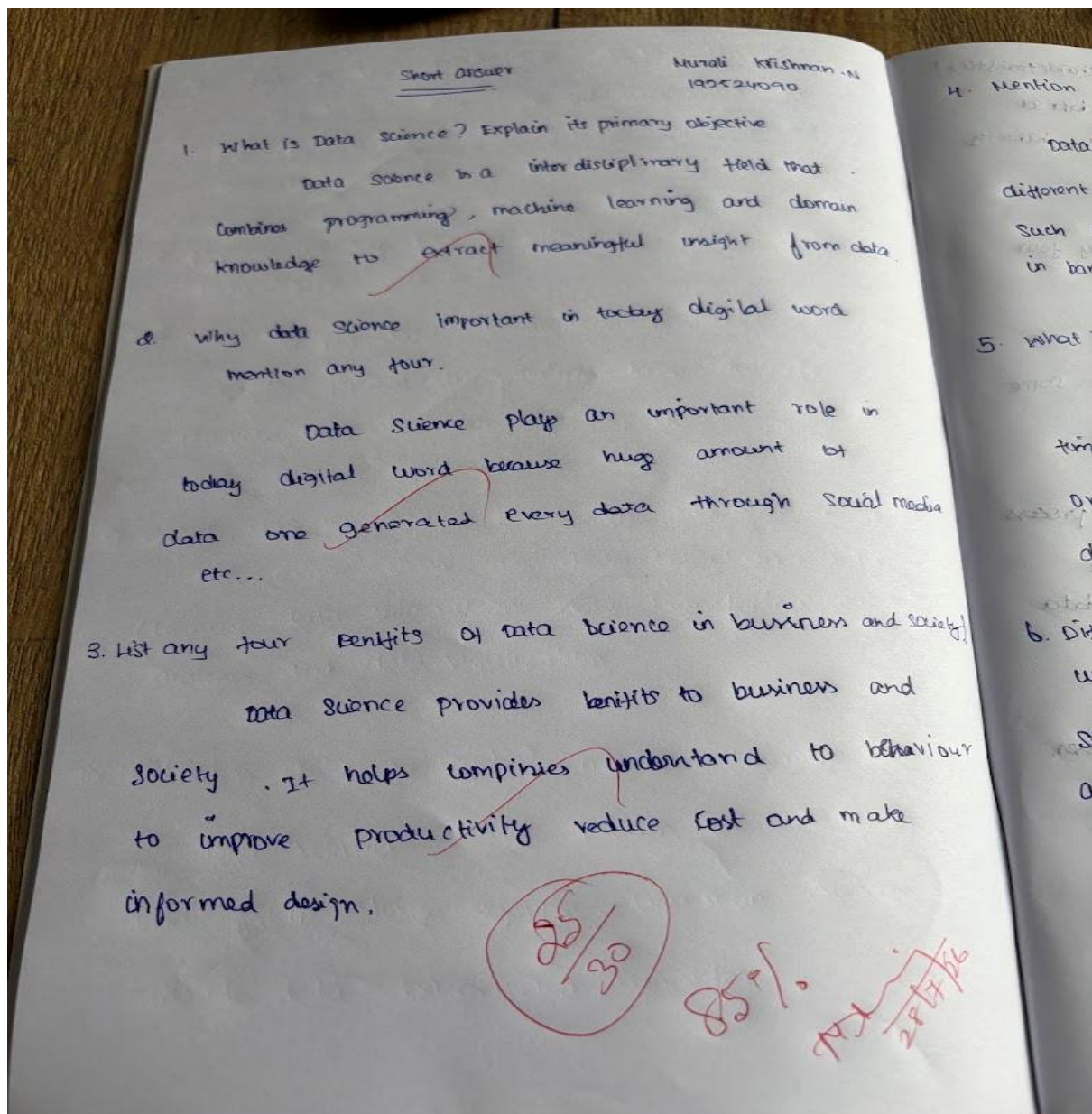


CO1-AT1-Short Answer Test

Slot D - DSA0405 - Fundamentals of Data Science



4. Mention four world application of data science?

Data science has many application across different industries. It is recommendation system such as Netflix and Amazon, fraud detection in banking, disease prediction in healthcare.

5. What are the different facets of data? Explain each.

The main facets of data structure, source, type, time, quality and usage, structure refer to the

organization of data science, source indicates where data comes and type describe the nature of data values.

6. Differentiate between structured, semi-structured and unstructured data.

Structured data organized in row and columns, such as database. Semi-structured data contains tag, json and xml. Unstructured data contains image, video, email

7. What is big data? state its key characteristics!

Big data is very large and complex data set. Its five characteristics are volume, velocity, variety, veracity and value.

8. Explain the big data ecosystem. Name any four technologies used in it.

The big data ecosystem consists of tools used to process and analyze large datasets. Some technologies are Hadoop, Spark.

9. What are the major phases of data science process?

The major phases are data collection, data cleaning, data integration, data transformation, data analysis and data visualization.

10. What is data retrieval? mention common sources.

Data retrieval is the process of obtaining data from storage system. Common sources

include data bases and websites.

11. Why data collection is important in Data Science?

Data collection because accurate and relevant data leads to reliable analysis and better decision.

12. What is data cleaning? List any four techniques.

Data cleaning is the process of correcting error in data. Common techniques include removing duplicate, handling missing value.

13. What is data integration? What is its necessity?

Data integration combines data from different sources into single dataset.

14. Explain data transformation with suitable example.

Data transformation convert data into suitable format for analysis.

15. What is data analysis? mention any four techniques.

Data analysis is the process of examining data to extract useful information.